

Again, the given number is divisible by 11.

So, $(x + 3 + 2 + 2 + 3) - (y + 1 + 2 + 4) = x - y + 3$ is either 0 or 11.

$\therefore (x - y + 3 = 0 \text{ or } x - y + 3 = 11) \Rightarrow (y - x = 3 \text{ or } x - y = 8)$

Now, $(y + x = 1 \text{ and } y - x = 3) \Rightarrow y = 2 \text{ and } x = -1$.

$$(y + x = 1 \text{ and } x - y = 8) \Rightarrow x = \frac{9}{2}.$$

$$(y + x = 10 \text{ and } y - x = 3) \Rightarrow y = \frac{13}{2}.$$

$$(y + x = 10 \text{ and } x - y = 8) \Rightarrow x = 9 \text{ and } y = 1.$$

Thus, $x = 9, y = 1$. So, the required number is 342221319.

235. 37×3 is divisible by 7

$\Rightarrow (7 \times 3 - 3)$ is either 0 or divisible by 7

$\Rightarrow 7 \times 0$ is divisible by 7.

$\Rightarrow X = 0$ or $X = 7$.

236. $225 = 9 \times 25$, where 9 and 25 are co-primes.

So, a number is divisible by 225 if it is divisible by both 9 and 25.

Given number is divisible by 25, only if $7q$ is divisible by 25, i.e. if $q = 5$.

Sum of digits of given number $= (8 + 7 + 6 + p + 3 + 7 + 5) = 36 + p$, which must be divisible by 9.

This is possible if $p = 0$.

Hence, $p = 0, q = 5$.

237. Since 774958A96B is divisible by 8, so the number 96B must be divisible by 8. So, $B = 0$, as 960 is divisible by 8. Now, 774958A960 is divisible by 9. So, $(55 + A)$ must be divisible by 9.

This happens when $A = 8$. Hence, $A = 8$ and $B = 0$.

238. $132 = 11 \times 3 \times 4$. So, the required number must be divisible by 3, 4 and 11.

264 $\rightarrow$ divisible by 3, 4 and 11.

396 $\rightarrow$ divisible by 3, 4 and 11.

462 $\rightarrow$ not divisible by 4.

792 $\rightarrow$ divisible by 3, 4 and 11.

968 $\rightarrow$ not divisible by 3

2178 $\rightarrow$ not divisible by 4

5184 $\rightarrow$ not divisible by 11.

6336 $\rightarrow$ divisible by 3, 4 and 11.

Hence, out of the given numbers 4 are divisible by 3, 4 and 11.

239. Let $3x + 7y = 11k$. Then, $y = \frac{(11k - 3x)}{7}$.

$$\text{Then, } 5x - 3y = 5x - \frac{3(11k - 3x)}{7} = \frac{35x - 33k + 9x}{7}$$

$$= \frac{44x - 33k}{7} = \frac{11(4x - 3k)}{7}, \text{ which is divisible by 11.}$$

240. $(1^3 - 1) = 0, (2^3 - 2) = 6, (3^3 - 3) = 24, (4^3 - 4) = 60$ and so on, each one of which is divisible by 6.

241. Let $a = 2k + 1$ & $b = 2m + 1$. Then,

$$\begin{aligned} a^4 - b^4 &= (a^2 - b^2)(a^2 + b^2) = (a + b)(a - b)(a^2 + b^2) \\ &= (2k + 2m + 2)(2k - 2m)(4k^2 + 4m^2 + 2 + 4k + 4m) \\ &= 8(k + m + 1)(k - m)(2k^2 + 2m^2 + 1 + 2k + 2m), \end{aligned}$$

which is divisible by 8.

242. Let the two consecutive integers be a and $(a + 1)$. Then, $(a + 1)^2 - a^2 = a^2 + 1 + 2a - a^2 = (2a + 1) = (a + a + 1) =$ sum of given integers.

243. $6n^2 + 6n = 6n(n + 1)$

$$(n = 1 \Rightarrow 6n^2 + 6n = 12), (n = 2 \Rightarrow 6n^2 + 6n = 36), (n = 3 \Rightarrow 6n^2 + 6n = 72),$$

each one of which is divisible by 6 and 12 both.

244. Let the ten's digit be x and the unit's digit be y . Then, $(10x + y) - (10y + x) = 9x - 9y = 9(x - y)$, which is divisible by 9.

245. Let the ten's digit be x and the unit's digit be y .

Then, $(10x + y) + (10y + x) = 11(x + y)$, which is divisible by 11.

246. Let $P = n(n + 1)(n + 2)(n + 3)$. Then, $n = 1$ gives $P = (1 \times 2 \times 3 \times 4) = 24$.

Hence, the required number is 24.

247. Let $N = n^2(n^2 - 1) = n^2(n - 1)(n + 1)$.

$$\text{Then, } n = 2 \Rightarrow N = 2^2 \times (2 - 1) \times (2 + 1) = (4 \times 1 \times 3) = 12.$$

Hence, the required number is 12.

248. Let $n = (2m + 1)$. Then,

$$N = n(n^2 - 1) = n(n - 1)(n + 1) = (2m + 1)(2m)(2m + 2) = 4m(m + 1)(2m + 1).$$

$$\text{Now, } m = 1 \Rightarrow N = (4 \times 1 \times 2 \times 3) = 24.$$

So, N is always divisible by 24.

249. Let the hundreds, tens and unit digits be x, y and z respectively.

$$\text{Then, } (100x + 10y + z) - (x + y + z) = 99x + 9y = 9(11x + y).$$

So, the resulting number is divisible by 9.

250. Let the required number be x . Then $\frac{11x + 11}{13} = a$ a whole number.

So, $(11x + 11)$ must be divisible by 13. By hit and trial, we get $x = 12$.

Hence, the smallest original number is 12.

251. Let the required product be $n(n + 1)(n + 2)$. Then,

$$n = 1 \Rightarrow n(n + 1)(n + 2) = (1 \times 2 \times 3) = 6.$$

$$n = 2 \Rightarrow n(n + 1)(n + 2) = (2 \times 3 \times 4) = 24.$$

$$n = 3 \Rightarrow n(n + 1)(n + 2) = (3 \times 4 \times 5) = 60.$$

So, each such product is divisible by 6.

252. Let the required odd numbers be $n, (n + 2)$ and $(n + 4)$.

Then $n + (n + 2) + (n + 4) = 3n + 6 = 3(n + 2)$, which is always divisible by 3.

But $n = 1 \Rightarrow 3(n + 2) = 3 \times 3 = 9$ which is not divisible by any of 2, 5 and 6.

Hence only II is true.

253. Three consecutive multiples of 3 are $3m, 3(m + 1)$ and $3(m + 2)$.

$$\text{Their product} = 3m \times 3(m + 1) \times 3(m + 2) = 27 \times m \times (m + 1) \times (m + 2).$$

$$\text{Putting } m = 1, \text{ this product is } (27 \times 1 \times 2 \times 3) = 162.$$

So, this product is always divisible by 162.

254. $p = 5 \Rightarrow (p^2 - 1) = (25 - 1) = 24$, which is divisible by 24.

$$p = 7 \Rightarrow (p^2 - 1) = (49 - 1) = 48, \text{ which is divisible by 24.}$$

$$p = 11 \Rightarrow (p^2 - 1) = (121 - 1) = 120, \text{ which is divisible by 24.}$$

Hence, $(p^2 - 1)$ is always divisible by 24.

255. Let the two consecutive odd integers be $(2m + 1)$ and $(2m + 3)$.

$$\text{Then, } (2m + 3)^2 - (2m + 1)^2 = [(2m + 3) + (2m + 1)][(2m + 3) - (2m + 1)] = (4m + 4) \times 2 = 8m + 8 = 8(m + 1), \text{ which is always divisible by 8.}$$